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$ whoami
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BENJAMIN_NIGJEH

> Chief Machine Learning Scientist | Computational Proteomics & AI

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cat summary.md

Computational proteomics scientist with 20+ years of interdisciplinary experience spanning computational biology, bioinformatics, scientific software engineering, proteomics, mass spectrometry, and pharmaceutical drug discovery. Designs and deploys AI-native computational platforms integrating machine learning, probabilistic modeling, graph algorithms, LLMs, and foundation models with classical bioinformatics to automate complex scientific workflows. Currently Chief ML Scientist at ProteoBio (ProteoAgent Server) and AI/ML Lead Scientist at Northwestern's Proteomics Center of Excellence; prior AI-enabled drug-discovery work at ActivX Biosciences, Vertex Pharmaceuticals, and Exelixis.

ls ./experience/

[ProteoBio, Inc.]

Chief Machine Learning Scientist

Dec 2025 - Present

// Lead AI/ML strategy; architect & principal developer of ProteoAgent Server.

- > Architected ProteoAgent Server: supervisor orchestration, expert AI agents, workflow planning, persistent memory, reporting, cloud deployment, interactive UIs.
- > Built ML frameworks across knowledge harvesting, dataset generation, MS signal processing, isotope deconvolution, feature discovery, protein ID, characterization, quantitation.
- > Developed the ProteoFormer Pipeline – containerized deep learning + probabilistic inference + graph algorithms + bioinformatics toolkits.
- > Shipped alpha release: automated mzML QC, protein ID, quantitative analysis, integrated reporting, multi-agent reasoning.

[Northwestern University – Proteomics Center of Excellence]

AI/ML Lead Scientist

Mar 2024 - Present

// Direct AI/ML methods for top-down proteomics and large-scale MS data analysis.

- > Founded & led the AI/ML Study Group: deep learning, protein language models, graph-based learning, foundation models, LLMs for proteomics.
- > Designed proteoform-discovery algorithms fusing MS1 scans, isotope envelopes, precursor ions, chromatographic evidence, MS/MS IDs.
- > Built IDNet (contrastive learning for MS/MS ID) and Graph-Based Proteoform Inference (GPI).
- > Led large-scale top-down proteomics studies: neurodegeneration, plasma proteomics, clinical biomarker discovery.

[ActivX Biosciences, Vertex Pharmaceuticals & Exelixis, Inc.]

Senior Scientist – Computational Proteomics, Bioinformatics & AI

2018 - 2024

// Data generation & computational solutions across small molecules, ADCs, gene editing, LNP mRNAs.

- > Built ML pipelines for proteomics/chemoproteomics/structural biology; retrained Prosit-derived spectrum-prediction models.
- > Designed MS workflows for PTMs, sequence variants, covalent binding, target engagement; built PK/PD & biomarker platforms.
- > Supported programs behind FDA-approved therapeutics: TRIKAFTA(R), KALYDECO(R), ORKAMBI(R), SYMDEKO(R), suzetrigine (VX-548).

[SUNY Buffalo & University of Washington School of Medicine]

Postdoctoral Research Scientist — Computational Proteomics & Quantitative MS*2011 — 2018*

- > Applied ML to DIA proteomics for biomarker discovery/disease classification; built pancreatic-cancer early-detection models.
- > Built DIA/SRM/PRM/label-free LC-MS workflows quantifying 3,500+ plasma proteins/glycoproteins from clinical cohorts.
- > Identified circulating galectin-3 binding protein as a candidate biomarker; published in Analytical Chemistry, JPR, and related journals.

cat education.yaml**University of Groningen****Ph.D., Analytical Biochemistry — 2006-2011**

// Dissertation: LC-MS methodologies for quantitative proteomics & pharmaceutical metabolism (hybrid electrochemistry-MS). Supported by AstraZeneca.

Sharif University of Technology**B.Sc./M.Sc., Materials Science & Engineering — 2000-2006****Self-Directed Postgraduate Study****CS, AI & Software Engineering — 2018-Present**

// MIT OpenCourseWare, O'Reilly, Manning, MIT Press: AI Engineering, LLMs, Agentic Systems, RAG, MLOps, Docker, Kubernetes, FastAPI, Distributed Systems.

cat skills.json**"ai_ml":**

[LLMs, Agentic/Multi-Agent Systems, Foundation Models, Protein Language Models, Deep Learning, Graph Learning, Bayesian Inference, XAI]

"proteomics":

[Top-Down/Bottom-Up, Native MS, DIA/DDA, Proteoform Informatics, Biomarker Discovery, Multi-Omics, Chemoproteomics]

"algorithms":

[Signal Processing, Isotope Deconvolution, Graph Algorithms, Statistical Modeling, Optimization, Clustering, Classification]

"software":

[Python, R, C/C++, Java, JavaScript, FastAPI, Docker, Kubernetes, Git, CI/CD, Linux, NGINX]

"cloud_infra":

[Distributed Systems, HPC, Data Engineering, MLOps, Containerized Workflows, Scalable AI Infra]

"ms_platforms":

[Orbitrap, UHMR Orbitrap, TIMS-TOF, Triple Quad, TOF, Ion Trap, LC-MS/MS]

"proteomics_sw":

[Skyline, Proteome Discoverer, ProSight PD, TDPortal, TopPIC, TopFD, UniDec, MaxQuant, PEAKS, pyOpenMS]

"data_science":

[Pandas, NumPy, Matplotlib, Plotly, Statistical Analysis, Dashboarding]